R and Tidyverse Basics

DATA SCIENCE BOOTCAMP

NASHVILLE SOFTWARE SCHOOL

R Introduction

Developed in New Zealand and first released in the mid-1990's. Version 1.0.0 was released in 2000.

An implementation of the S programming language (developed by Bell Labs in the 1970s)

Heavily used in academic research and statistical analysis

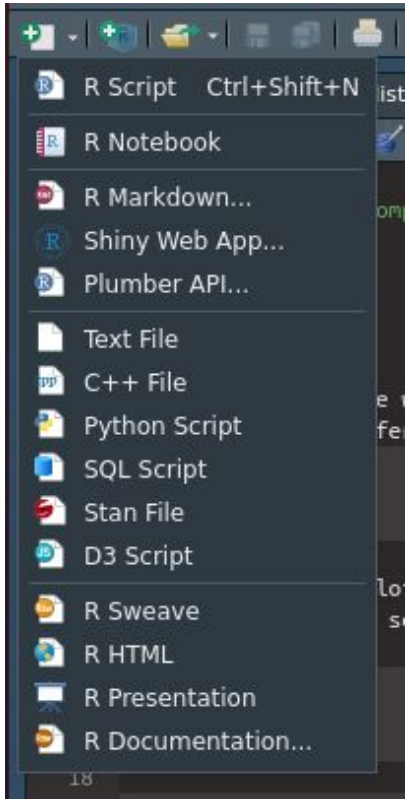
Recently efforts to broaden appeal and modernize the language – tidyverse

Types of R Files

The two most common types of files you will work with in R and R Scripts (.R) and R Notebooks (.Rmd).

The major difference is that code is separated into chunks separated by text in an R Notebook, and the result can be knit to a pdf or HTML file. This is the type of file we'll normally use.

When working with Shiny, we'll use R Scripts.



Key Differences From Python

- Indexing begins at 1.
- Assignment is made with the assignment operator:

```
x <- 7
```

- Note: An equals sign also works for assignment (x = 7).

Vectors in R

- Homogeneous (all entries must be the same datatype).
- Created using the c operator (c for “combine”)

```
x <- c(1,2,3)
```

- You can also assign names to the elements of a vector:

```
x <- c(“first” = 1, “second” = 2, “third” = 3)
```

- You can access element of a vector either by index (remember, indexing starts at 1) or by name.

```
x[1] or x[‘first’]
```

- **Think:** numpy array or pandas Series.

Lists in R

- Heterogeneous - can have mixed datatypes.
- Created using the list constructor

```
x <- list(1,2,3)
```

- You can also assign names to the elements of a list:

```
x <- list("first" = 1, "second" = 2, "third" = 3)
```

- You can access element of a vector either by index (remember, indexing starts at 1) or by name. When referencing the name, you can use either brackets or $

```
x[1] or x['first'] or x$first
```

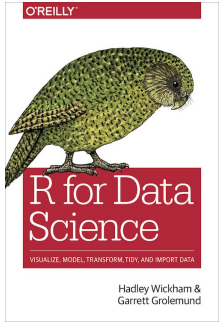
- **Think:** Python lists or dictionaries

What is the Tidyverse?

The tidyverse is an opinionated collection of R packages designed for data science.

All packages share an underlying design philosophy, grammar, and data structures.

Learn more about the tidyverse: R for Data Science - https://r4ds.hadley.nz/



Tibbles

The basic data structure of the tidyverse is the tibble.

A tibble is a slightly improved version of R's data frame.

See [this blog post](#) for more information about the differences.

```
> flights
# A tibble: 336,776 x 19
    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time arr_delay carrier
   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>     <dbl> <chr>
 1  2013     1     1      517            515         2      830            819        11 UA
 2  2013     1     1      533            529         4      850            830        20 UA
 3  2013     1     1      542            540         2      923            850        33 AA
 4  2013     1     1      544            545        -1     1004           1022       -18 B6
 5  2013     1     1      554            600        -6      812            837       -25 DL
 6  2013     1     1      554            558        -4      740            728        12 UA
 7  2013     1     1      555            600        -5      913            854        19 B6
 8  2013     1     1      557            600        -3      709            723       -14 EV
 9  2013     1     1      557            600        -3      838            846        -8 B6
10  2013     1     1      558            600        -2      753            745         8 AA
# ... with 336,766 more rows, and 9 more variables: flight <int>, tailnum <chr>, origin <chr>,
#   dest <chr>, air_time <dbl>, distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dttm>
```

Finding Help

When Googling make sure to include “tidyverse” in your search.

There are also several really helpful cheatsheets here:

https://posit.co/resources/cheatsheets/

Pipes

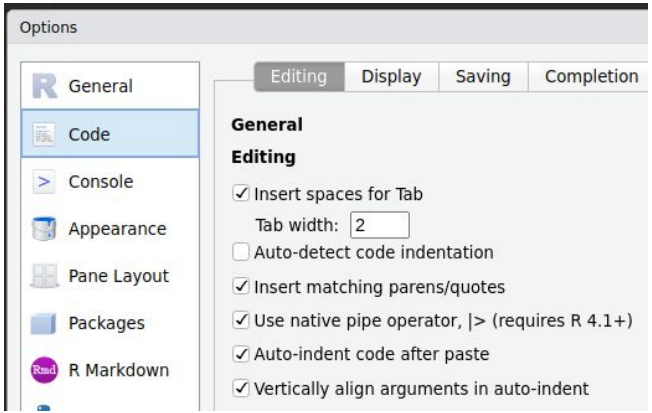
The pipe operator, from the magrittr library, is %>%.

As of R 4.1.0, there is also a pipe operator in base R, |>.

In RStudio, it can be inserted using the shortcut Ctrl + Shift + M or Cmd + Shift + M. (By default, this will use the magrittr pipe, but can be changed in the RStudio settings).

Takes the output of one statement and makes it the input of the next statement.

Makes your code more readable and more writable.



Pipes

Compare

```
my_data <- do_one_last_thing(
        do_something_else(
                do_something(data_in)
                )
        )
```

versus

```
my_data <- data_in |>
        do_something() |>
        do_something_else() |>
        do_one_last_thing()
```

tidyverse vs pandas

pandas	tidyverse
.loc / .iloc / .query / [...]	

tidyverse vs pandas

pandas	tidyverse
.loc / .iloc / .query / [...]	filter

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange
calculated column/.assign	

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange
calculated column/.assign	mutate

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange
calculated column/.assign	mutate
.groupby	

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange
calculated column/.assign	mutate
.groupby	group_by

tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange
calculated column/.assign	mutate
.groupby	group_by
aggregate function (.mean, .median, etc.)	

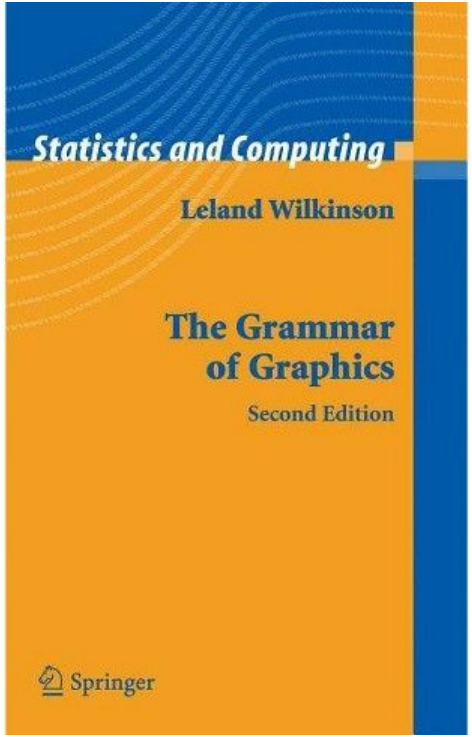
tidyverse vs pandas

pandas	**tidyverse**
.loc / .iloc / .query / [...]	filter
sort_values	arrange
calculated column/.assign	mutate
.groupby	group_by
aggregate function (.mean, .median, etc.)	summarize

ggplot

The ggplot2 library takes the “grammar of graphics” approach to data visualization.

Any plot can be built up from “layers”.



ggplot

The top three are needed for any visualization.

Element	Description
Data	The data-set being plotted.
Aesthetics	The scales onto which we *map* our data.
Geometries	The visual elements used for our data.
Themes	All non-data ink.
Statistics	Representations of our data to aid understanding.
Coordinates	The space on which the data will be plotted.
Facets	Plotting small multiples.

borrowed from *Introduction to Data Visualization with ggplot2* DataCamp course

ggplot - aesthethics

Aesthetic	Description
x	X axis position
y	Y axis position
fill	Fill color
color	Color of points, outlines of other geoms
size	Area or radius of points, thickness of lines

Aesthetics start with which variables go on which axes but then also include which other variables you want to visualize through color, size, etc.

Borrowed from *Introduction to Data Visualization with ggplot2* DataCamp course

ggplot - geometries

See https://ggplot2.tidyverse.org/reference/ for a more complete list.

- geom_point: scatterplot
- geom_bar/geom_col: bar chart
- geom_histogram
- geom_boxplot